

Distribuzioni

Sia Ω aperto $\subseteq \mathbb{R}^n$

$$C_0^\infty(\Omega) := \left\{ f: \Omega \rightarrow \mathbb{R} \text{ di classe } C^\infty, \right. \\ \left. \uparrow \text{ aventi supporto compatto in } \Omega \right\}$$

spazio vettoriale. $\left[f, g \in C_0^\infty(\Omega), \alpha, \beta \in \mathbb{R} \Rightarrow \right. \\ \left. \alpha f + \beta g \in C_0^\infty(\Omega) \right]$

Def. convergente in $C_0^\infty(\Omega)$ il nulla

Diciamo che $\{\varphi_n\} \subseteq C_0^\infty(\Omega)$ converge a $\checkmark$ 0 in $C_0^\infty(\Omega)$ se

1) $\exists K$ (indep. da n) tale che

$$\text{Supp}(\varphi_n) \subseteq K \quad \forall n > \nu$$

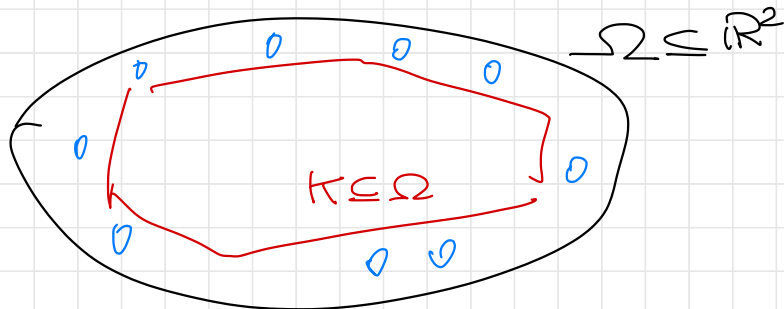
2) $\varphi_n \rightarrow 0$ uniformemente con tutte le derivate

$$(D^\alpha \varphi_n \rightarrow 0 \text{ unif. su } K)$$

$$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k)$$

$$D^\alpha \varphi = \frac{\partial^{\alpha_1 + \alpha_2 + \dots + \alpha_k} \varphi}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \dots \partial x_k^{\alpha_k}}$$

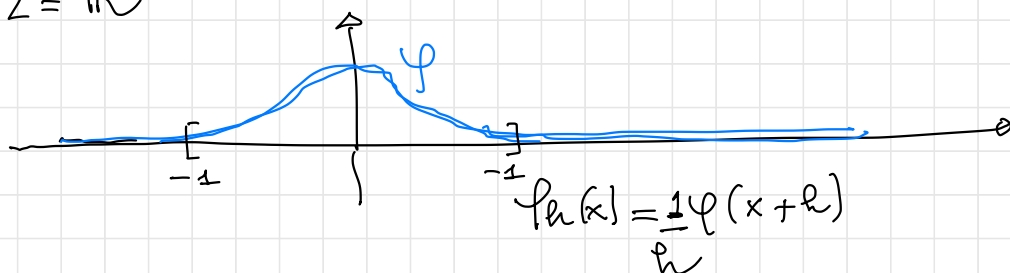
$$D^{2,1,4} \varphi = \frac{\partial^7 \varphi}{\partial x_1^2 \partial x_2^1 \partial x_3^4}$$



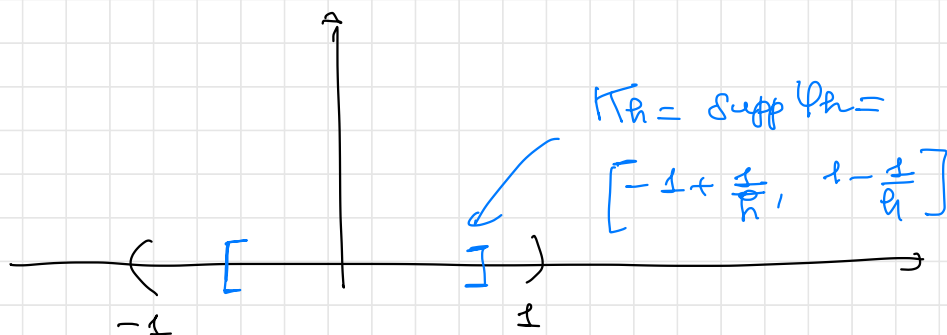
Es.

- Casi in cui 1) non è verificata

$$\Omega = \mathbb{R}$$



$$\Omega = (-1, 1)$$



- $\varphi_h \rightarrow \varphi$ in $\mathcal{C}_0^\infty(\Omega)$ $\stackrel{\text{def.}}{\iff} \varphi_h - \varphi \rightarrow 0$ in $\mathcal{C}_0^\infty(\Omega)$
- Qs convergente non proviene da una norma su $\mathcal{C}_0^\infty(\Omega)$

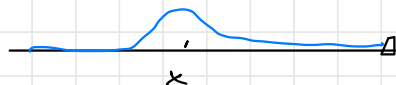
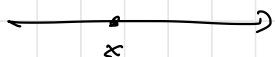
(X ||. || su $C_0^\infty(\Omega)$) tale che

$$\varphi_n \rightarrow \varphi \text{ in } C_0^\infty(\Omega) \text{ (seconda def)} \Leftrightarrow \|\varphi_n - \varphi\| \rightarrow 0$$

Def. $C_0^\infty(\Omega)$ munito della convergenza introdotta si indica con $\mathcal{D}(\Omega)$ e si chiama Spazio delle funzioni test.

Idea:

$$u \mapsto u(x) \rightsquigarrow u \mapsto \int_{\Omega} u \varphi$$



Lemma $u \in L^1(\Omega)$; $\int_{\Omega} u \varphi = 0 \quad \forall \varphi \in C_0^\infty(\Omega)$
 $\Rightarrow u = 0$ q.o. su Ω

$u_1, u_2 \in L^1(\Omega)$; $\int_{\Omega} u_1 \varphi = \int_{\Omega} u_2 \varphi \quad \forall \varphi \in C_0^\infty(\Omega)$
 $\Rightarrow u_1 = u_2$ (q.o. Ω).

Def. Una DISTRIBUZIONE su Ω

è un operatore $T: \mathcal{D}(\Omega) \rightarrow \mathbb{R}$
tale che:
 $\uparrow$
sp. vettoriale

(i) T è lineare

$$(T(\alpha f + \beta g) = \alpha T(f) + \beta T(g) \quad \forall f, g \in \mathcal{D}(\Omega) \\ \forall \alpha, \beta \in \mathbb{R})$$

(ii) T è continuo

rispetto alla convergenza su $\mathcal{D}(\Omega)$

ovvero,

$$\boxed{\varphi_h \rightarrow \varphi \text{ in } \mathcal{D}(\Omega) \Rightarrow T(\varphi_h) \rightarrow T(\varphi)}$$

Dim.

- N.B. $\mathcal{D}(\Omega)$ non è sp. vett. normato
 $\Rightarrow$ una distrib. non è un op. lineare
su uno sp. vett. normato
- T è lineare $\Rightarrow T$ è continuo se e solo
è continuo in 0

Per verificare T continuo, basta

$$\boxed{\varphi_h \rightarrow 0 \text{ in } \mathcal{D}(\Omega) \Rightarrow T(\varphi_h) \rightarrow 0}$$

Def. Lo spazio delle distribuzioni su Ω si indica con $\mathcal{D}'(\Omega)$



è uno spazio vettoriale

ovvero $T_1, T_2 \in \mathcal{D}'(\Omega), \alpha, \beta \in \mathbb{R}$

$$\Rightarrow (\alpha T_1 + \beta T_2) \in \mathcal{D}'(\Omega)$$

verifica elementare

(i) linearità

(ii) continuità

$$[\mathcal{D}(\Omega), \mathcal{D}'(\Omega)]$$

Esempi

i) Sia $u \in L^1(\mathbb{R})$.

Associo a u una distribuzione T_u definita da:

$$\langle u, \varphi \rangle \rightarrow T_u(\varphi) := \underbrace{\int_{\Omega} u \varphi}_{\substack{\uparrow \\ \mathbb{R}}} \quad \forall \varphi \in \mathcal{D}(\mathbb{R})$$

• T_u è ben definito

$$\left| \int_{\Omega} u \varphi \right| \leq \int_{\Omega} |u \varphi| \leq \max_K |\varphi| \cdot \underbrace{\int_K |u|}_{< +\infty}$$

$K = \text{supp } \varphi$

$$\text{ii) } T_u(\alpha_1 \varphi_1 + \alpha_2 \varphi_2) \stackrel{?}{=} \alpha_1 T_u(\varphi_1) + \alpha_2 T_u(\varphi_2)$$

$$\downarrow$$
$$\int_{\Omega} u(\alpha_1 \varphi_1 + \alpha_2 \varphi_2)$$

$$\downarrow$$
$$\alpha_1 \int_{\Omega} u \varphi_1 + \alpha_2 \int_{\Omega} u \varphi_2$$

$$\int_{\Omega} \alpha_1 u \varphi_1 + \alpha_2 u \varphi_2 \quad \leftarrow \quad \text{per linearità dell'integrale}$$

(ii) Sia $\{\varphi_n\} \rightarrow 0$ in $\mathcal{D}(\mathbb{R})$

È vero che $T_u(\varphi_n) \rightarrow 0$?

$$T_u(\varphi) = \int_{\Omega} u \varphi \rightarrow 0 ?$$

$$\left| \int_{\Omega} u \varphi \right| \leq \int_{\Omega} |u \varphi| \leq \underbrace{\max_K |\varphi|}_{\downarrow 0} \cdot \underbrace{\int_{\Omega} |u|}_{\substack{\leq K \\ + \infty}}$$

osservazioni

• Notazione:

$$u \in L^1(\Omega) \rightsquigarrow T_u \in \mathcal{D}'(\Omega)$$

$$T_u(\varphi) = \langle T_u, \varphi \rangle_{(\mathcal{D}'(\Omega), \mathcal{D}(\Omega))} = \langle u, \varphi \rangle_{\mathcal{D}', \mathcal{D}}$$

• lemma: siano $u_1, u_2 \in L^1(\Omega)$

$$u_1 = u_2 \text{ q.o.} \iff T_{u_1} = T_{u_2} \text{ in } \mathcal{D}'(\Omega)$$

ovvero:

$$\int_{\Omega} u_1 \varphi = \int_{\Omega} u_2 \varphi \quad \forall \varphi \in \mathcal{D}(\Omega)$$

$$\left(u = u_1 - u_2 = 0 \text{ q.o.} \iff \int_{\Omega} u \varphi = 0 \quad \forall \varphi \in \mathcal{D}(\Omega) \right)$$

- Abbiamo solo usato, per associare

$$u \rightsquigarrow T_u \in \mathcal{D}'(\Omega)$$

la seguente proprietà

$$u \in L^1(K) \quad \forall K \text{ compatto di } \Omega$$

È più debole rispetto a $u \in L^1(\Omega)$

Esempio $\Omega = (0, 1)$ $u(x) = \frac{1}{x} \notin L^1(0, 1)$

Ma $\forall K \text{ compatto} \subseteq (0, 1)$, $\frac{1}{x} \in L^1(K)$.

$$L^1_{loc}(\Omega) := \left\{ u: \Omega \rightarrow \mathbb{R} : \int_K |u| < +\infty \right. \\ \left. \forall K \text{ compatto} \subseteq \Omega \right\}.$$

$$L^1(\Omega) \subsetneq L^1_{loc}(\Omega)$$

perché se $u \in L^1(\Omega) \Rightarrow u \in L^1(K)$
 $\forall K \text{ compatto} \subseteq \Omega$

$L^1_{loc}(\Omega) \ni u \rightsquigarrow T_u \in \mathcal{D}'(\Omega)$

• In particolare:

$$u \in L^p(\Omega) \quad p \in [1, +\infty].$$

$$L^p(\Omega) \ni u \rightsquigarrow T_u \in \mathcal{D}'(\Omega)$$

$$T_u(\varphi) = \int_{\Omega} u \varphi \quad \forall \varphi \in \mathcal{D}(\Omega).$$

$$L^p(\Omega) \subseteq L^1_{loc}(\Omega)$$

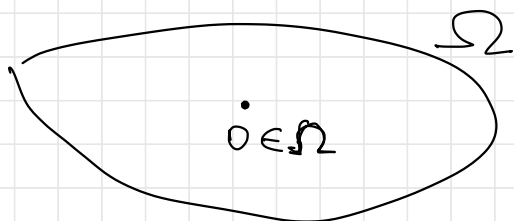
Se $u \in L^p(\Omega)$, K compatto $\subseteq \Omega$

$$\int_K |u| < +\infty \quad \text{perché}$$

$$u \in L^p(\Omega) \Rightarrow u \in L^p(K) \subseteq L^1(K)$$

$\uparrow$
perché $|K| < +\infty$

2) Delta di Dirac in $0 \in \Omega$



Def. $\delta_0 \in \mathcal{D}'(\Omega)$ definita da:

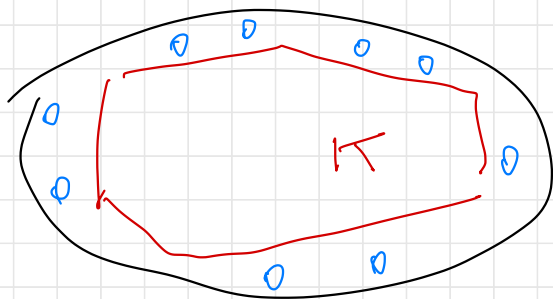
$$\delta_0(\varphi) = \langle \delta_0, \varphi \rangle_{\mathcal{D}', \mathcal{D}} := \varphi(0) \quad \forall \varphi \in \mathcal{D}(\Omega)$$

Verifica:

$$(i) \quad \delta_0(\alpha_1 \varphi_1 + \alpha_2 \varphi_2) \stackrel{?}{=} \alpha_1 \delta_0(\varphi_1) + \alpha_2 \delta_0(\varphi_2)$$

$$\begin{array}{ccc} \downarrow & & \downarrow \\ (\alpha_1 \varphi_1 + \alpha_2 \varphi_2)(0) & \stackrel{=?}{\longleftrightarrow} & \alpha_1 \varphi_1(0) + \alpha_2 \varphi_2(0) \\ \parallel & & \\ \alpha_1 \varphi_1(0) + \alpha_2 \varphi_2(0) & & \end{array}$$

$$(ii) \quad \varphi_n \rightarrow 0 \text{ in } \mathcal{D}(\Omega) \stackrel{?}{\Rightarrow} \underbrace{\delta_0(\varphi_n)}_{\varphi_n''(0)} \rightarrow 0$$



$$0 \notin K \Rightarrow \varphi_n(0) = 0 \quad \forall n$$

$$0 \in K \Rightarrow \varphi_n(0) \rightarrow 0$$

perché su K ho
conv. uniforme a 0.

Claim: $\nexists u \in L^1_{loc}(\Omega)$ tale che $T_u = \delta_0 \leftarrow \begin{matrix} \uparrow \\ \text{in } \mathcal{D}'(\Omega) \end{matrix}$
 ovvero $\nexists u \in L^1_{loc}(\Omega)$ tale che
 $\langle T_u, \varphi \rangle = \langle \delta_0, \varphi \rangle \quad \boxed{\forall \varphi \in \mathcal{D}(\Omega)}$
 $\int_{\Omega} u \varphi \quad \varphi(0)$

$n=1$ $\Omega = \mathbb{R}$

• Passo preliminare $\boxed{\varphi \in C_0^\infty(\mathbb{R} \setminus \{0\}) \subseteq C_0^\infty(\mathbb{R})}$

$\text{---} \quad [K] \quad \bullet \quad [K] \quad \text{---}$

$$\int_{\mathbb{R}} u \varphi = \int_{\underbrace{\mathbb{R} \setminus \{0\}}_{\Omega}} u \varphi = 0$$

Lemma

se $u \in L^1(\Omega)$ e tale che $\int_{\Omega} u \varphi = 0 \quad \forall \varphi \in C_0^\infty(\Omega)$
 $\Rightarrow u=0$ q.o. su Ω

Quindi:

$u=0$ q.o. su $\mathbb{R} \setminus \{0\}$.

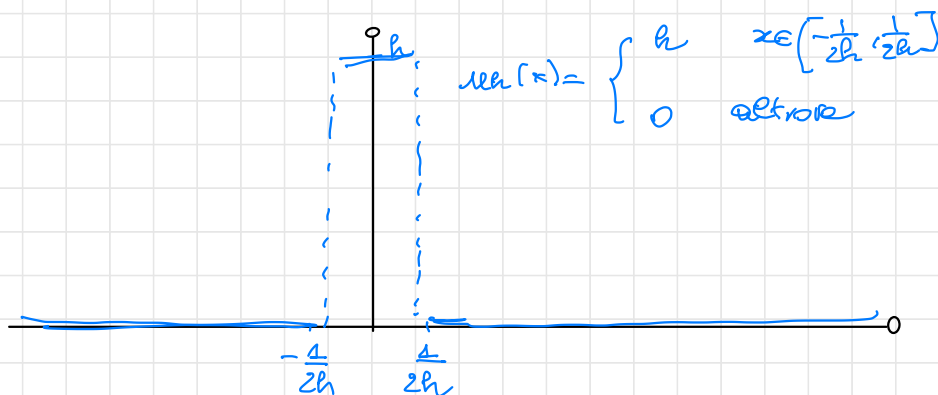
che questo equivale a dire

$\boxed{u=0 \text{ q.o. su } \mathbb{R}}$

- Posso prendere anche $\bar{\varphi} \in C_0^\infty(\mathbb{R})$ tale che $\bar{\varphi}(0) = 1$

$$\int_{\mathbb{R}} u \bar{\varphi} \rightarrow \begin{cases} 0 & \text{perché } (u\bar{\varphi}) = 0 \text{ q.o. su } \mathbb{R} \\ \bar{\varphi}(0) = 1 \end{cases}$$

□



Per ogni h , $u_h \in L^1(\mathbb{R})$. $\int_{\mathbb{R}} |u_h| = \frac{1}{h} \cdot h = 1$

$u_h \rightarrow 0$ q.o. su $\mathbb{R}$.

Quindi $\nexists \lim u_h$ in $L^1(\mathbb{R})$.

$$\begin{aligned} T_{u_h}(\varphi) &= \int_{\mathbb{R}} \varphi u_h = \int_{-\frac{1}{2h}}^{\frac{1}{2h}} h \varphi \\ &= \frac{1}{\frac{1}{h}} \int_{-\frac{1}{2h}}^{\frac{1}{2h}} \varphi = \int_{-\frac{1}{2h}}^{\frac{1}{2h}} \varphi \rightarrow \varphi(0) \\ &\quad \delta_0(\varphi) \end{aligned}$$

$$T_{\text{lin}}(\varphi) \rightarrow \delta(\varphi) \quad \forall \varphi \in \mathcal{D}_0^\infty(\Omega)$$

$$\Uparrow$$

$$T_{\text{lin}} \rightarrow \delta \quad \text{in } \mathcal{D}'(\Omega)$$

Def. $T_n \xrightarrow{n} T \stackrel{\text{def}}{\iff} T_n(\varphi) \rightarrow T(\varphi) \quad \forall \varphi \in \mathcal{D}(\Omega).$

$\uparrow$
convergence in $\mathcal{D}'(\Omega).$

Derivazione di distributivi

"Idea" $\int_0^1 u' \varphi = - \int_0^1 u \varphi' + \cancel{u \varphi \Big|_0^1}$

Def. ($\Omega \subseteq \mathbb{R}$). Date $T \in \mathcal{D}'(\Omega)$, definisco $T' \in \mathcal{D}'(\Omega)$ date da

$$\langle T', \varphi \rangle \stackrel{\text{def}}{=} - \langle T, \varphi' \rangle \quad \forall \varphi \in \mathcal{D}(\Omega)$$

Verifica:

i) $\mathcal{L}$ lineare?

$$\begin{aligned} \langle T', \alpha_1 \varphi_1 + \alpha_2 \varphi_2 \rangle &\stackrel{?}{=} \alpha_1 \langle T', \varphi_1 \rangle + \alpha_2 \langle T', \varphi_2 \rangle \\ &\stackrel{\text{"def}}{=} - \langle T, (\alpha_1 \varphi_1 + \alpha_2 \varphi_2)' \rangle \\ &\stackrel{\text{"}}{=} - \langle T, \alpha_1 \varphi_1' + \alpha_2 \varphi_2' \rangle \\ &\stackrel{\text{"}}{=} - \alpha_1 \langle T, \varphi_1' \rangle - \alpha_2 \langle T, \varphi_2' \rangle \end{aligned}$$

" $\leftarrow T$ è lineare

$$\begin{aligned} \text{ii) } \underbrace{\varphi_n \rightarrow 0 \text{ in } \mathcal{D}(\Omega)}_{\Downarrow \text{ segue dalla def}} &\stackrel{?}{\Rightarrow} \langle T', \varphi_n \rangle \rightarrow 0 \\ &\stackrel{\text{"}}{=} - \langle T, \varphi_n' \rangle \rightarrow 0 \text{ vero} \\ &\quad \text{perché } \varphi_n' \rightarrow 0 \text{ in } \mathcal{D}(\Omega) \end{aligned}$$

• Def. Se $T = T_u$ con $u \in C^1(\Omega)$

($\Rightarrow$ sia u che u' sono in $L^1_{loc}(\Omega)$)

(perché continue su $\Omega \Rightarrow$ sono integrabili sui compatti di Ω)

$\Rightarrow$ ha senso considerare $T_u, T_{u'}$

$$\boxed{(T_u)' = T_{u'}}$$

perché $\forall \varphi \in \mathcal{D}(\Omega)$

$$\left\{ \begin{aligned} \langle (T_u)', \varphi \rangle &\stackrel{\text{per def}}{=} - \langle T_u, \varphi' \rangle = \boxed{- \int_{\Omega} u \varphi'} \\ \langle T_{u'}, \varphi \rangle &\stackrel{\text{per def}}{=} \boxed{\int_{\Omega} u' \varphi} \end{aligned} \right. \begin{aligned} &\Longleftrightarrow \text{int per} \\ &\text{parti} \\ &(+ \varphi = 0 \text{ su } \partial\Omega) \end{aligned}$$

Dato $u \in C^1(\Omega) \subseteq L^1_{loc}(\Omega)$

$$\langle \underset{T_u}{u}, \varphi \rangle$$

$$\langle \underset{\varphi}{u'}, \varphi \rangle = - \langle u, \varphi' \rangle \quad \forall u \in \mathcal{D}(\Omega)$$

Exempi:

1) $u(x) = |x|$ in $\mathcal{D}'(-1,1)$

$$\langle (T_{|x|})', \varphi \rangle \stackrel{\text{per def}}{=} - \langle T_{|x|}, \varphi' \rangle = \quad \forall \varphi \in \mathcal{D}(-1,1).$$

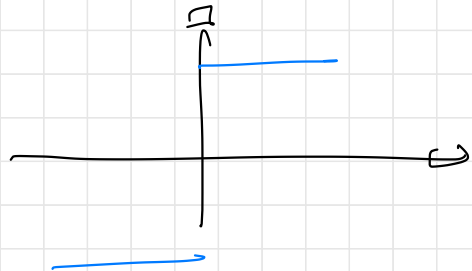
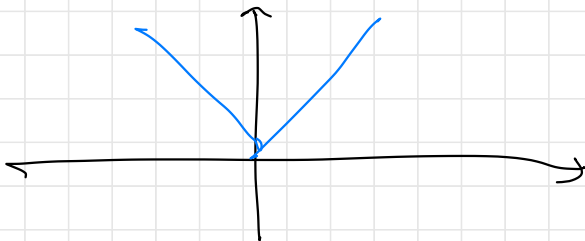
$$= - \int_{-1}^1 |x| \varphi' = - \int_0^1 x \varphi' + \int_{-1}^0 x \varphi'$$

$$= \int_0^1 \varphi + \cancel{x \varphi(x)} \Big|_0^1 - \int_{-1}^0 \varphi(x) + \cancel{x \varphi(x)} \Big|_{-1}^0$$

$$= \int_{-1}^1 [\text{sign}(x)] \varphi(x) = \langle T_{\text{sign}(x)}, \varphi \rangle$$

$\begin{array}{cc} \swarrow & \searrow \\ 1 & -1 \\ x > 0 & x < 0 \end{array}$

$$\boxed{(|x|)' = \text{sign}(x) \quad \text{in } \mathcal{D}'(-1,1)}$$



$$2) \quad u(x) = \text{sign}(x) \quad \forall \varphi \in \mathcal{D}(-1, 1).$$

$$\langle u', \varphi \rangle \stackrel{\text{per def}}{=} -\langle u, \varphi' \rangle = -\int_{-1}^1 \text{sign}(x) \varphi' dx$$

$$= -\int_0^1 \varphi' dx + \int_{-1}^0 \varphi' dx$$

$$= -\cancel{\varphi(1)} + \varphi(0) + \varphi(0) - \cancel{\varphi(-1)}.$$

$$= 2\varphi(0) = 2\delta_0, \varphi$$

$$\boxed{(\text{sign } x)' = 2\delta_0}$$

$$3) \quad T = \delta_0 \quad T' = ?$$

$$\langle (\delta_0)', \varphi \rangle \stackrel{\text{per def}}{=} -\langle \delta_0, \varphi' \rangle \stackrel{\text{per def}}{=} -\varphi(0)$$

$$\uparrow \quad \uparrow$$

$$\text{per def} \quad \text{per def}$$

$$\mathcal{D}(\delta_0)' \quad \mathcal{D}(\delta_0)$$

$$\forall \varphi \in \mathcal{D}(\mathbb{R})$$

generalizzazioni

• $n=1$ Date $T \in \mathcal{D}'(\Omega)$, $\forall k \in \mathbb{N}$
 $T^{(k)} \in \mathcal{D}'(\Omega)$ è definita da

$$\langle T^{(k)}, \varphi \rangle \stackrel{\text{def}}{=} (-1)^k \langle T, \varphi^{(k)} \rangle \quad \forall \varphi \in \mathcal{D}(\Omega)$$

$$\begin{aligned} \langle T'', \varphi \rangle &= -\langle T', \varphi' \rangle \quad \forall \varphi \in \mathcal{D}(\Omega) \\ &= -(-\langle T, \varphi'' \rangle) \\ &= \langle T, \varphi'' \rangle \quad \forall \varphi \in \mathcal{D}(\Omega) \end{aligned}$$

$$\langle T', \varphi \rangle = -\langle T, \varphi' \rangle$$

• $n \geq 1$ Note $T \in \mathcal{D}'(\Omega)$ ($\Omega \text{ op. } \subseteq \mathbb{R}^n$)

$$\langle \bigoplus^{\alpha} T, \varphi \rangle \stackrel{\text{def}}{=} (-1)^{|\alpha|} \langle T, \bigoplus^{\alpha} \varphi \rangle$$

$\forall \varphi \in \mathcal{D}(\Omega)$

α multi index

$$(\alpha_1, \dots, \alpha_n)$$

$$|\alpha| = \alpha_1 + \dots + \alpha_n$$